

Complex numbers



- 1 What type of number can be written in the form $a + bi$?
 - 2 Complete the following statement:
An imaginary number is a complex number that can be written in the form bi .
 - 3 Consider the imaginary number i .
 - a What is the number i equal to?
 - b What does i^2 equal?
 - 4 Consider the complex number $-i\sqrt{10}$.
 - a Write the real part.
 - b Write the imaginary part.
 - c State the type of number this is.
 - 5 Write down the complex number that has:
 - a A real part of 0 and an imaginary part of $\sqrt{5}$.
 - b A real part of $\sqrt{14}$ and an imaginary part of 0.
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Operations with complex numbers

- 6 Rewrite $9 + \sqrt{-4}$ in the form $a + bi$.
- 7 Laura wants to evaluate $-\sqrt{-16}$ and writes out the following working out:

$$\begin{aligned} -\sqrt{-16} &= -(-4) \\ &= 4 \end{aligned}$$

Is this working out correct? If not, write the correct answer.

- 8 Consider the statement:

$$\frac{6 \pm 36i}{2} = 3 \pm 36i$$

Is this statement correct? If not, simplify the statement correctly.

9 Express the following in terms of i :



a $-\sqrt{-100}$

b $\sqrt{-18}$

c $9 - \sqrt{-64}$

d $2 + \sqrt{-25}$

10 Express $i\sqrt{-64}$ without a negative radicand.

11 Evaluate:

a $\sqrt{-19} \times \sqrt{-19}$

b $-\sqrt{-5} \times \sqrt{-5}$

c $\frac{\sqrt{-10}}{\sqrt{-2}}$

d $\frac{\sqrt{-5}}{\sqrt{-45}}$

12 Evaluate:

a $(3 + 6i) + (7 + 3i)$

b $(9 + 2i) - (6 - 7i)$

c $(-1 + 5i) - 2$

d $(6 - 8i) - 3i$

e $(6.5 + 7.9i) + (-9.8 - 5.3i)$

f $(-4 + 3i) + (-1 + 5i) + (-6 + 8i)$

g $(-6 - i) - (7 - 5i) - (-2 + 9i)$

h $(2 + 3i) - (-6 + 8i) + (4 - i)$

i $\left(\frac{1}{2} - \frac{i}{4}\right) + \left(\frac{2}{3} + \frac{i}{5}\right)$

13 Evaluate:

a $-6(3 - 5i)$

b $7i(3 - 2i)$

c $\sqrt{10}i(8 + \sqrt{10}i)$

d $(5 - 7i)(8 + 8i)$

e $(1 + 8i)(1 - 8i)$

f $(2 + 5i)(5i - 2)$

g $(2 - 5i)^2$

h $\left(\frac{4}{5} + \frac{3}{5}i\right)\left(\frac{1}{4} + \frac{3}{4}i\right)$

i $2i(3 - i)^2$

j $-2i(4 - 3i)^2$

14 Consider the imaginary number i .

a Evaluate:

i i^2

ii i^4

iii i^6

iv i^8

b Consider the expression i^n . State whether the following values of n would result in i^n being equal to 1.

i 16

ii 22

iii 4

iv 14

15 Simplify:



a $2i^7$

b vi^{10}

c $4i^{10}$

d $10i^8$

e $(3i)^3$

f $(2i)^4$

g $5i^2 + 7i^3 - 8i^4$

h $8i^3 - 7i^4 + 6i^5$

Complex conjugates

16 The numbers $3 + 4i$ and $3 - 4i$, which differ only in the sign of the imaginary parts, are called what?

17 The product of a complex number and its conjugate is always what type of number?

18 Multiply each of the following numbers by its conjugate:

a $2 + 4i$

b $-6 - \sqrt{3}i$

c $1 + 5i$

d $-4 + 3i$

e $-4 - i$

f $\sqrt{2} + i$

g $4 - \sqrt{5}i$

h $\sqrt{2} + \sqrt{3}i$

19 Evaluate:

a $\frac{4 - 6i}{2}$

b $\frac{5 + 2i}{i}$

c $\frac{-5}{6i}$

d $\frac{-5 + 2i}{i}$

e $\frac{4 + 6i}{1 + i}$

f $\frac{7 + 4i}{2 + 3i}$

g $\frac{4 + 7i}{2 + i}$

h $\frac{2 - 3i}{2 + 3i}$

i $\frac{2 - i}{1 + 4i}$

j $\frac{2 + 2i}{2 - i}$

20 Consider the expression $5 + \sqrt{6}i$:

a Find the sum of the expression and its conjugate.

b Find the product of the expression and its conjugate.

21 Evaluate:

a $\frac{3}{1 + i} + \frac{2}{4 - i}$

b $\frac{8}{i} + \frac{7}{i^2} - \frac{2}{i^3}$

22 Write $-\frac{1}{5}i^3(\sqrt{-16} - 25) - 4i^2$ in simplest $a + bi$ form.

Application



- 23** Complex numbers are used in electronics to describe the current in an electric circuit. Ohm's law relates the current in a circuit, I , in amperes, the voltage of the circuit, E , in volts, and the resistance of the circuit, R , in ohms, by the formula $E = IR$. Using the formula given above, find E , the voltage of a circuit, if $I = 9 - 4i$ amperes and $R = 1 + 6i$ ohms.